

SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

Group Name: Last minute

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1. Introduction:

This system analysis document highlights the technical scope, architectural boundaries, and design-related decisions behind the development of StitchFlow V2. It serves to transition the product from a high-risk autonomous agent to a secure, human-in-the-loop decision-support copilot.

1.1. Purpose:

The system analysis document highlights the technical scope and design-related decision behind the project.

So, the key element of system that has been covered are as followed:

1. Architecture:

- a. A Redaction-First, human-in-the-loop (HITL) engine integrated with Google Workspace.

2. Data Flows:

- a. The movement of messy, unstructured operational inputs through pre-processing, GLM-4.5 reasoning, and final structured output into Google Sheets and Gmail.

3. Model Process:

- a. A multi-step prompt reasoning chain performing entity extraction, ambiguity detection, and policy screening before a human approves execution.

4. Role of Reference:

- a. This document acts as the primary blueprint for the development and QA teams to maintain architectural integrity during the hackathon.

1.2. Background:

Event and operations teams handle highly fragmented, sensitive, and unstructured inputs such as PDFs, chat messages, emails, and spreadsheet notes. Manual coordination is error-prone, while fully autonomous AI processing creates unacceptable business risks.

1. Previous Version:

- a. **Autonomous Execution Risk:** V1 assumed AI could autonomously execute actions, leading to risks of hallucination and overcommitment .
- b. **Privacy & Compliance Blindspot:** V1 sent raw operational data directly to the LLM, increasing privacy exposure .

- c. **Hallucinated Authority:** V1 made silent compliance judgments without providing evidence or confidence scores .
- 2. Changes in Major Architectural Components:
 - a. **Redaction-First Architecture:** PII is deterministically masked before reaching the model.
 - b. **Decision-Support Focus:** The system drafts and suggests; human users must explicitly approve execution.
 - c. **Explicit Stateful Workflow Logic:** Every item moves through strictly tracked states (e.g., Needs Clarification, Ready for Review) ensuring recoverability.
- 3. New capabilities that are only introduced in this project:
 - a. **Evidence-Based Policy Screening:** V2 shifts to a screening model where the AI flags potential issues by providing a confidence score and directly quoting source evidence from the input documents.
 - b. **Reviewable Spreadsheet Integration:** V2 transforms AI reasoning into structured, operationally manageable Google Sheets rows. Each row represents an atomic workflow item (Item ID, Source, Extracted Entities, Confidence, Policy Flag) waiting for human approval.

1.3. Target Stakeholder

Stakeholders	Roles	Expectations
Ops Manager	Reviewing extracted entities and approving system actions.	Systematic approval workflow via Google Sheets.
Compliance Officer	Auditing and reviewing uploaded documents against policy libraries.	Evidence-based flags with cited source material.
Event Coordinator	Inputting multiple past fragmented data (emails, chats, PDFs).	Expects a clear, color-coded UI distinguishing precedent clauses from AI recommendations.
Dev / QA Team	Maintaining the state machine and GLM prompts.	Strict JSON schema validation and retry logic.

2. System Architecture & Design

2.1 High Level Architecture

2.1.1 Overview:

Type	Details
System	Web-based Operations Dashboard & Google Workspace Integration
Architecture	Hybrid HITL Engine: Deterministic Redaction + Probabilistic Reasoning

StitchFlow is primarily structured as a Stateful Human-in-the-Loop (HITL) Engine utilizing a hybrid architecture. The system interacts with users through multiple interfaces, including a web-based Policy Screening Dashboard, a T&C Comparison UI, and Google Workspace integrations like Google Sheets and Gmail. These interfaces communicate with a backend orchestrator that separates deterministic validation and system state management from the probabilistic reasoning powered by Z.AI’s GLM-4.5. The core processing flow relies on distinct interconnected modules: a Pre-AI Redaction Engine, a multi-step GLM Reasoning chain, and a Dynamic Task Orchestration service. All workflow items are ultimately managed by a Stateful Workflow Engine that explicitly tracks item states and logs transitions across the dashboard and external APIs.

2.1.2 LLM as Service Layer

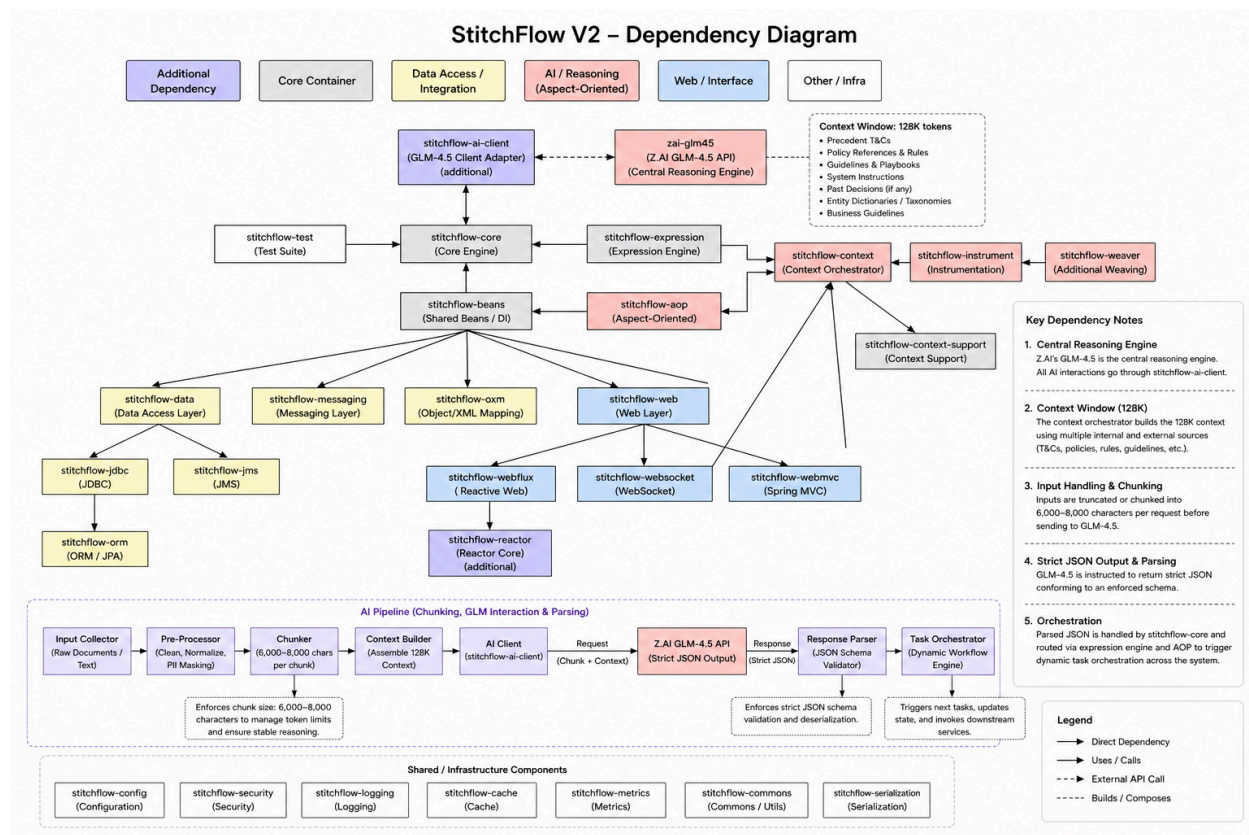
GLM-5.1 is the central reasoning engine, providing Agentic Consistency and a 128K Context Window.

2.1.3 Dependency Diagram

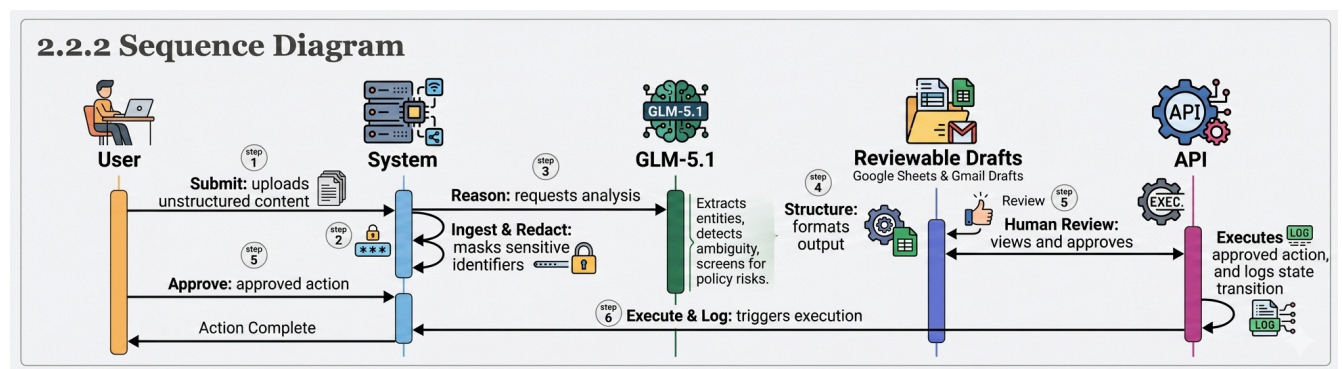
StitchFlow V2 utilizes Z.AI's GLM-4.5 as the central reasoning engine.

- **Context Window:** Leverages GLM's 128K context to evaluate large precedent T&Cs and policy references simultaneously.
- **Input Handling & Chunking:** Inputs are truncated or chunked at 6,000-8,000 characters per workflow request to maintain stable reasoning and manage token constraints .

- **Parsing:** The system enforces a strict JSON schema output from GLM, which the backend parses to trigger dynamic task orchestration.



2.2.2. Sequence Diagram



2.2 Technological Stack

2.2.1 Frontend

React.js / Next.js (Dashboard and T&C Simulation UI).

2.2.2 Backend

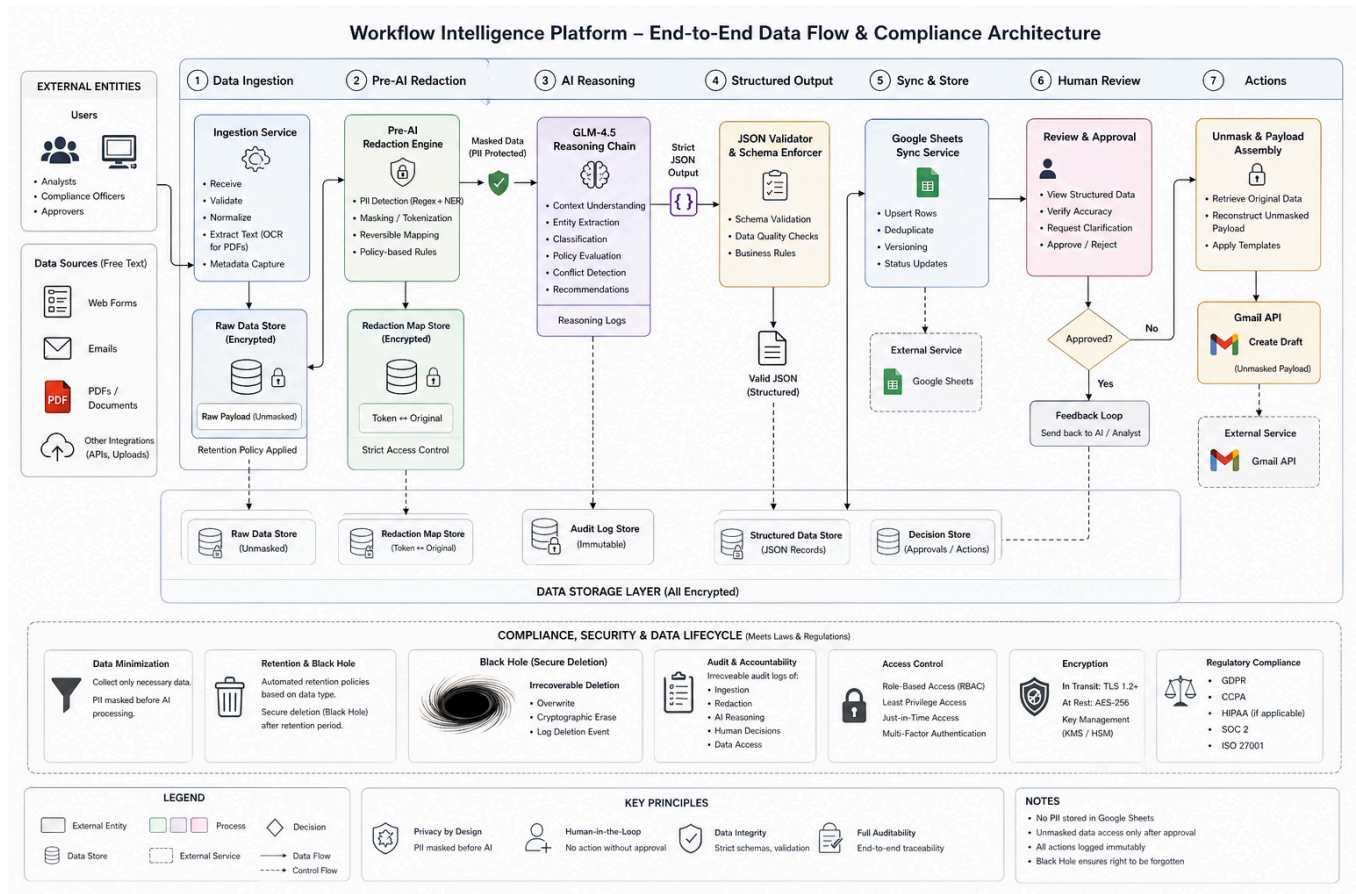
Node.js / FastAPI (Workflow orchestration and state management).

2.2.3 Database

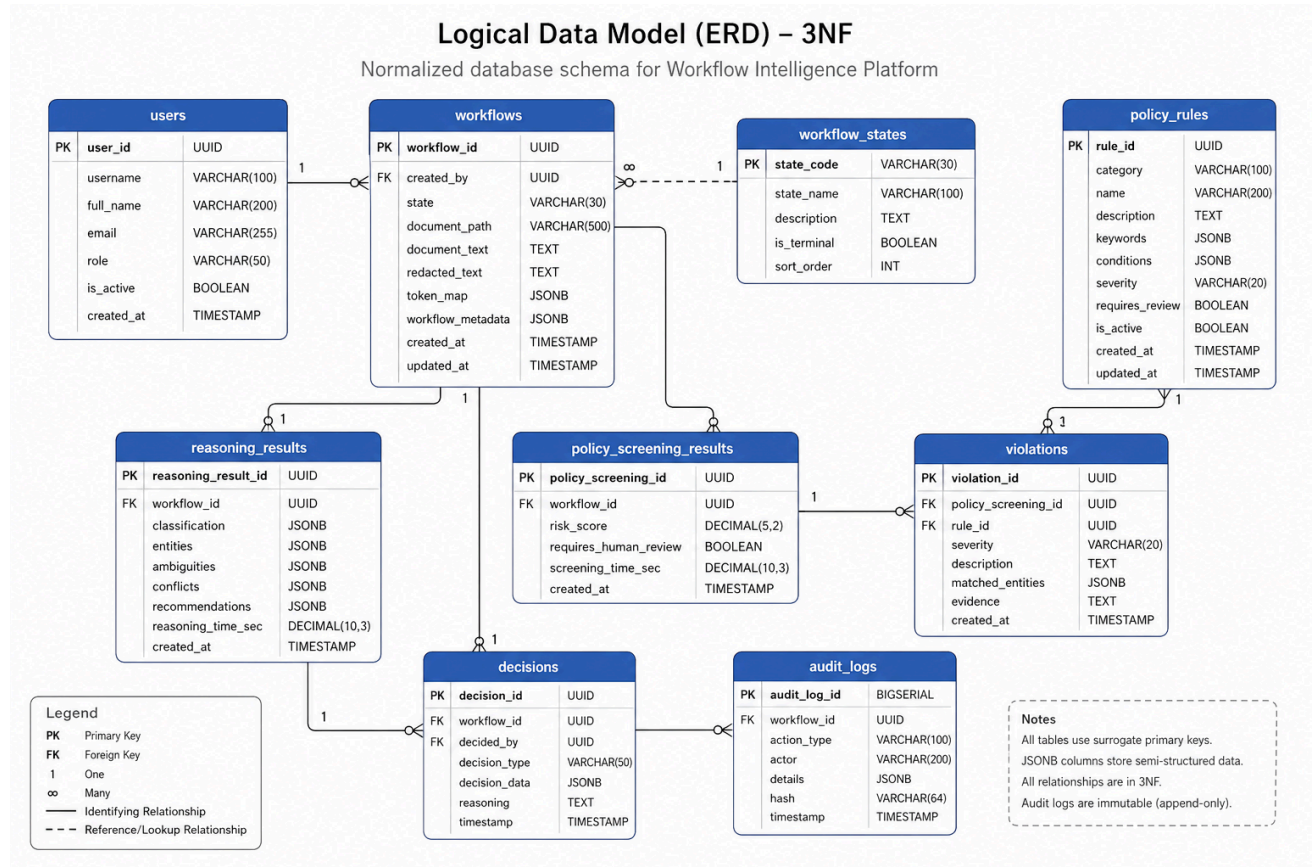
SQLite (Storing normalized workflow states and audit logs).

2.3 Key Data Flows

2.3.1 Data Flow Diagram (DFD)



2.3.2 Normalized Database Schema



3. Functional Requirements & Scope

3.1 Minimum Viable Product:

3.1.1

#	Features	Description
1	Ingestion & Redaction	File upload with automatic masking of sensitive data before reasoning.
2	Z.AI GLM-5.1 Reasoning	Multi-step prompt chain for extraction, contradiction checks, and next-action recommendation.
3	Google Sheets Sync	AI-generated rows for review and state management, acting as the operational dashboard.

4	Gmail API Integration	Draft creation for outbound communication, ensuring human sign-off.
5	Stateful Workflow Engine	System manages explicit item states (e.g., Needs Clarification, Ready for Review).

3.2 Non-Functional Requirements (NFRs)

Quality (e.g.)	Requirements (e.g.)	Implementation (e.g.)
Privacy	Zero PII exposure to LLM.	Implemented via the Pre-AI Redaction Engine masking data locally.
Reliability	No silent failures during API or model errors.	Schema validation, preserved partial states, and routing failed items to a "Retry" or "Escalated" queue.
Cost Efficiency	Must maintain financial feasibility for high-volume workflows.	Utilization of Z.AI context caching and prompt structuring to limit token consumption.
Auditability	All decisions, edits, and overrides must be traceable.	The system captures real human overrides and logs them into a distinct Audit Trail.

3.3 Out of Scope / Future Enhancements

- a. Fully autonomous messaging or execution without human approval.
- b. Automatic legal or regulatory certification.
- c. Full production-grade WhatsApp automation.
- d. Large-scale model fine-tuning and long-term predictive analytics .

4. Monitor, Evaluation, Assumptions & Dependencies

4.1 Technical Evaluation:

4.1.1 **Grayscale Rollout:** Core workflow paths (like Sheets syncing) will be deployed incrementally to test API limits.

4.1.2 **Emergency Rollback (ER):** If GLM outputs consistently fail schema validation, the system gracefully degrades into passive storage and rigid connector logic, bypassing the AI reasoning layer entirely.

4.1.3 Priority Matrix:

4.1.3.1 If hallucinated extraction conflicts with known rules, the system blocks final execution and defaults to the Needs Clarification state.

4.2 Monitoring:

4.2.1 The system continuously monitors the "Confidence Score" of the AI. Scores below acceptable thresholds trigger automatic routing to human review.

4.3 Assumptions:

- Multi-step workflow prompting is financially feasible under prototype budget constraints (~\$0.60 per 1M input tokens).
- Execution speed will be intentionally bottlenecked by human review cycles for high-impact steps.

4.4 External Dependencies:

Tools	Purpose	Risks
Z.AI GLM-5.1 API	Core reasoning, multi-step orchestration, and policy screening.	Misclassification or downtime.
Google Sheet API	Serves as the structured review UI and operational tracking state.	Spreadsheet Drift.

5. Project Management & Team Contributions

5.1 Project Timeline:

Day 1 - 4: Brainstorming idea and project planning

Day 5: Attend the talk and mentorship as well as assign the tasks

Day 6 - 9: Organise the meeting and collaborate to build the website

Day 10: Draft and finalise the report

5.2 Team Members Role:

Members	Roles
Liew Jian Wei	Project Leader
Damian Heng	Project Developer
Hon Chang Wei	Project Developer
Joyce Wong Tze Eng	Complete the report
Yong Pei Yu	Complete the report

5.3 Recommendations:

Implement caching for repeated system instructions and stable policy contexts to drastically reduce token costs and improve latency during the demo.